INCH-POUND

MIL-G-53133/2
30 August 1993

MILITARY SPECIFICATION SHEET

GENERATOR SETS, SKID MOUNTED, TACTICAL, QUIET,

DIESEL ENGINE DRIVEN, ALTERNATING CURRENT

5 KILOWATTS, 400 HERTZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-G-53133.

1. SCOPE

1.1 Scope. This specification sheet covers the detail requirements for tactical generator sets, size 5 kW, mode II (400 Hz), model MEP-812A, NSN 6115-01-274-7391.

2. APPLICABLE DOCUMENTS

2.1 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

| | |
|---|---|
| TA 30554-88-812 | - Generator Set, Tactical, Quiet, 5 kW, 400 Hz. |
| 30554-88-23000 | - Instructions, Preservation and Packing, 5 kW. |

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-FGM, Fort Belvoir, VA 22060-5606.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-G-53133 as specified herein.

AMSC N/A FSC 6115

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ADHESIVES

3.2 Description. The set shall be in accordance with TA 30554-88-812 and as specified herein.

3.3 Reliability. The minimum acceptable mean time between failure (MTBF) for the set shall be 442 hours with a 80 percent lower confidence limit.

3.4 Size and weight. The maximum operating set dimensions with control panel doors closed and the total operational weight shall not exceed the following:

| Parameter | Dimension |
|---|---|
| Length | 1280 mm (50.4 inches) |
| Width | 808 mm (31.8 inches) |
| Height | 920 mm (36.2 inches) |
| Weight | 413 kg (911 pounds) |

3.5 Fuel consumption. Fuel consumption, at all loads up to and including rated load, shall not be more than 2.65 ℓ/h (0.70 gallons per hour [GPH]) when the set is operating on referee grade diesel fuel conforming to MIL-F-46162.

3.6 Frequency regulation. The governing system shall maintain the frequency within 3 percent of rated frequency for all loads from no load up to and including rated load.

3.7 Frequency short-term stability (30 seconds). At every constant load from no load to rated load, the system shall maintain frequency within a bandwidth equal to 2 percent of rated frequency. The system shall not permit repetitive frequency variations, commonly called "hunting", even though within the allowable 2 percent band.

3.8 Frequency long-term stability (4 hours). At constant ambient temperature, constant barometric pressure, constant voltage, and any constant load from no load to rated load, the system shall maintain frequency within a bandwidth equal to 3 percent of rated frequency in a 4-hour operation period.

3.9 Frequency drift. With the regulation set at not more than 3 percent and with the set operating at constant load and voltage, a change in ambient temperature of up to 33 C° (60 F°) in an 8-hour period shall not cause the frequency to change by more than 2 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

3.10 Frequency transient performance. Following any sudden increase in load, including from no load to rated load, the governing system shall re-establish stable engine operating conditions within 3 seconds, and the maximum transient frequency change below the new steady-state frequency (undershoot) shall be not more than 3 percent of rated frequency. Following any sudden decrease in load, including from rated load to no load, the



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governing system shall re-establish stable engine operating condition within 3 seconds, and the maximum transient frequency change above the new steady-state frequency (overshoot) shall be not more than 4 percent of rated frequency.

3.11 Frequency adjustment. By means of the frequency adjust device, it shall be possible to adjust the frequency, when operating at any constant load, between 398.2 and 401.8 Hz. It shall not be possible to adjust the frequency to a value which activates the overspeed protective device.

3.12 Voltage waveform. The deviation factor and any single voltage harmonic for the line-to-line and line-to-neutral voltage for each voltage connection and load connection shall be not more than the following:

a. The deviation factor shall not exceed 6 percent in the single phase connections or 5 percent in the 3-phase, 120/208 volt connection.

b. Single frequency harmonics shall not exceed 3 percent in the single phase connections or 2 percent in the 3-phase, 120/208 volt connection.

In addition, there shall be no evident discontinuities, spikes, or notches in the waveform when viewed on an oscilloscope having a bandwidth of DC to 1.5 MHz and usable viewing screen of 8 x 10 cm. The oscilloscope gain shall be adjusted such that one cycle of voltage wave covers approximately the entire viewing screen.

3.13 Voltage unbalance. Maximum difference between line-to-line voltages shall be not more than 5 percent of rated voltage for the 120/208-volt connection under the conditions of a single phase, line-to-line unity pf-load of 25 percent rated current and no other load on the generator.

3.14 Phase balance voltage. With the set operating at rated voltage and frequency and no load, maximum difference in the three line-to-neutral voltages shall be not more than 1 percent of the rated line-to-neutral voltage when connected for 120/208-volt, 3-phase, 4 wire. The maximum difference between the voltage of the 120-volt windings of any phase shall not exceed 1 volt.

3.15 Voltage modulation. The voltage modulation of the output voltage for any load up to and including rated load shall not exceed 2.5 percent. This requirements shall apply to all line-to-line and line-to-neutral voltages.

3.16 Voltage regulation. The voltage regulation from no load to rated load and from rated load to no load shall be not more than 3 percent of the rated voltage for all voltage connections throughout the voltage ranges specified in 3.22.

3.17 Voltage short-term stability (30 seconds). At every constant load from no load to rated load, the voltage at the set terminals shall remain within a bandwidth equal to 2 percent of rated voltage.



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[illegible]: Excess conductor wire should be trimmed sufficiently close to provide [illegible]

3.18 Voltage long-term stability (4 hours). At constant ambient temperature, constant barometric pressure, constant frequency and any constant load from no load to rated load, the voltage shall remain within a bandwidth of 4 percent of rated voltage in a 4-hour operating period.

3.19 Voltage drift. With the set operating at constant load and frequency, a change in ambient temperature up to 33 C° (60 F°) in an 8-hour period shall not cause the voltage to change by more than 2 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

3.20 Voltage dip and rise for rated load. Performance of the set under transient conditions (as measured by a magnetic oscillograph) shall be as follows: With the set initially operating at rated frequency and rated voltage, and following any sudden change in load from no load to rated load, the instantaneous terminal voltage shall not dip more than 20 percent of rated voltage and shall reach stable conditions within 3 seconds. Overshoot or undershoot of the final voltage shall not exceed the initial voltage transient in amplitudes. The above requirements shall also apply when the load is suddenly changed from rated load to no load, except that the initial voltage transient shall involve a voltage rise not to exceed 20 percent of rated voltage.

3.21 Voltage dip for low power factor load. Performance of the set under transient conditions (as measured by a magnetic oscillograph) shall be as follows: With the set operating at no load and the corresponding voltage and frequency conditions, the output voltage of the set shall not dip more than 35 percent of the no load voltage when a balanced 3-phase 0.4 pf (or less) lagging static load of twice rated set current (1/2 per unit impedance) is suddenly applied to the set output terminals. The output voltage shall recover to a minimum of 95 percent of the no load voltage within 5 seconds and shall stabilize at or above this voltage.

3.22 Voltage adjustment range. The minimum voltage adjustment range shall be:

| Voltage Connection | Range |
|---|---|
| 120/208 V | 205 to 220 volts |
| 120 V | 114 to 126 volts |
| 240 V | 228 to 252 volts |

This voltage adjustment range shall apply for all loads between no load and rated kW, rated power factor at rated frequency under all environmental conditions. It shall not be possible to adjust the voltage of a value which activates the overvoltage protective device.

3.23 Endurance. The set shall operate for 3,000 hours without critical failure at all loads, continuous and intermittent, up to and including rated



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load under all of the individual or combination of conditions specified in 3.9.1 of the general specification.

3.24 Exciter field current. With the generator temperature stabilized at normal ambient conditions, the exciter field current shall not exceed 1.56 amperes for rated load at 400 Hz operation.

4. PACKAGING

4.1 Preservation, packing and marking. Preservation, packing and marking shall be in accordance with 30554-88-23000.

Custodians:
Army - ME
Navy - YD
Air Force - 99

Preparing activity:
Army - ME

Project 6115-0266

Review activities:
Navy - MC
DLA - GS



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